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Nottingham

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Computer Performance

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AY2025-26, Spring Semester
COMP1047: Systems and Architecture
Week 1



- **Timing Performance**
- **Other Performance Metrics**





Response Time and Throughput

- Response time (execution time)
 - The **total time** for the computer to complete a task
- Throughput (bandwidth)
 - The **number of tasks** completed per unit time
- How are response time and throughput affected?
 - Replacing the processor with a faster version?
 - Improves both response time and throughput
 - Adding more processors?
 - Improves throughput



- Response time
 - **Total time** to complete a task, including all aspects:
 - Disk accesses, memory accesses, I/O activities, CPU time, etc.
- CPU execution time (CPU time)
 - The **actual time the CPU spends** computing a specific task
 - Does not include time spent waiting for I/O or running other programs
 - Can be further divided into
 - User CPU time: the CPU time spent in a program itself
 - System CPU time: the CPU time spent in the OS performing tasks on behalf of the program



Relative Performance

- To maximize performance, we want to minimize execution time, then we can relate performance and execution time for a computer X as

$$\text{Performance}_X = 1 / \text{Execution_Time}_X$$

- Computer X is n times faster than computer Y, then their relative performance n is

$$\begin{aligned} n &= \text{Performance}_X / \text{Performance}_Y \\ &= \text{Execution_Time}_Y / \text{Execution_Time}_X \end{aligned}$$

- **Question: If Computer A runs a program in 10 secs and Computer B runs the same program in 15 secs, which one is faster? And by how much?**



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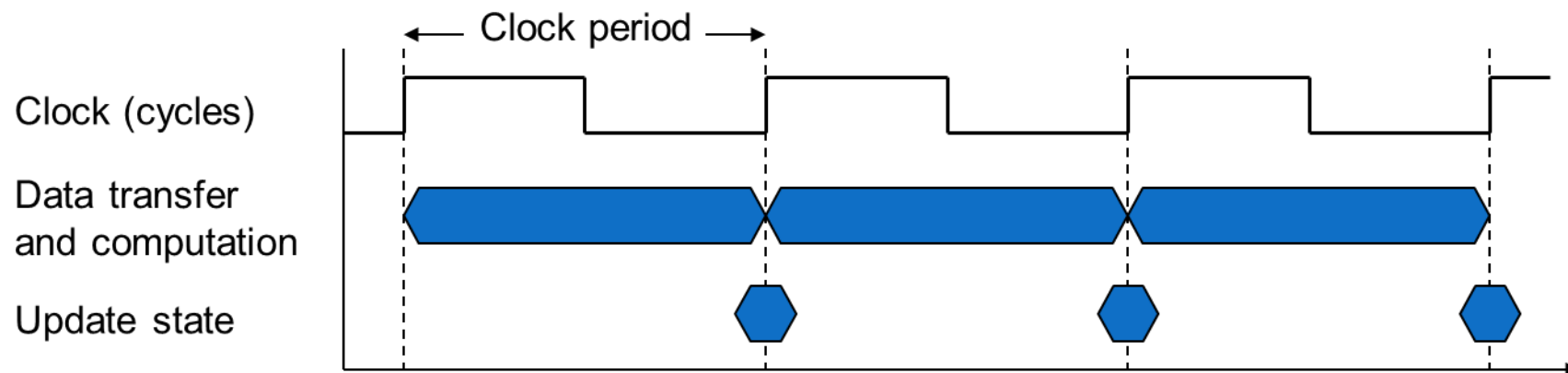
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- **Question: If Computer A runs a program in 10 secs and Computer B runs the same program in 15 secs, which one is faster? And by how much?**
- **Computer A is 1.5 times as fast as Computer B**
 - Or say “A is 50% faster than B”

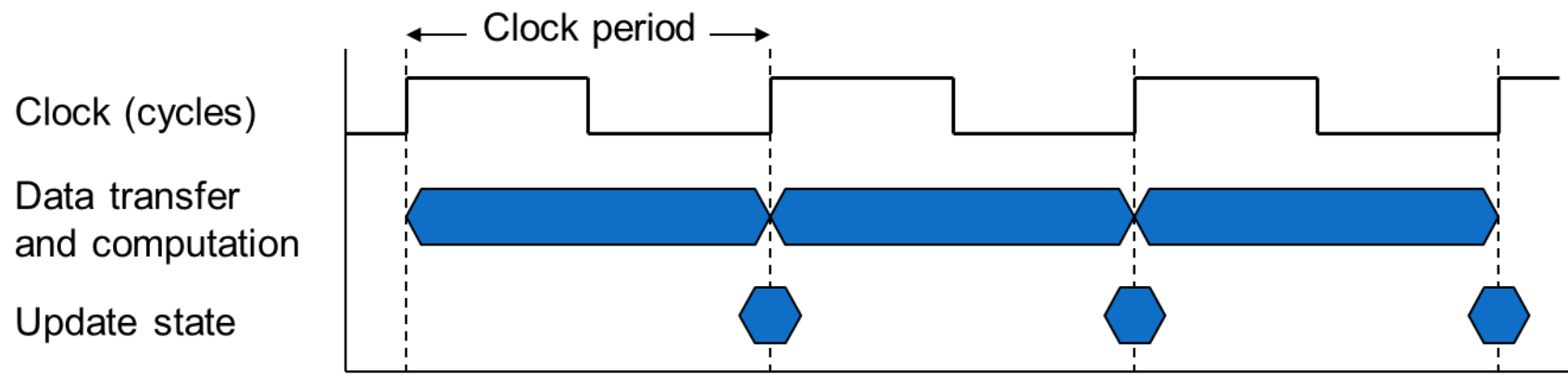
CPU Clocking

- Computers are constructed using a clock that determines when events take place in hardware
- The clock signal is produced by an external oscillator circuit that generates a consistent number of pulses each second in the form of a periodic square wave
- **One clock cycle (clock tick): the unit of the CPU clock. It must be constant as CPU clock runs.**
- **Clock period: the time spend to run one clock cycle.**



CPU Clocking

- **One clock cycle (clock tick):** the unit of the CPU clock. It must be constant as CPU clock runs.
 - e.g. $250\text{ps} = 0.25\text{ns} = 250 \times 10^{-12}\text{s}$ (ps: picosecond)
- **Clock rate (clock frequency):** cycles per second, which is the inverse of the clock period
 - e.g. $4.0\text{GHz} = 4000\text{MHz} = 4.0 \times 10^9\text{Hz}$
 - **Clock frequency = $1/\text{Clock period}$**





- A simple formula relates CPU clock cycles and CPU clock period to CPU time

$$\begin{aligned}\text{CPU Time} &= \# \text{CPU Clock Cycles} \times \text{Clock Period} \\ &= \frac{\# \text{CPU Clock Cycles}}{\text{Clock Rate}}\end{aligned}$$

- The performance can be improved by
 - Reducing number of clock cycles
 - Increasing clock rate
 - But they can not be altered arbitrarily – needs to adhere to circuit limitations.



CPU Time Example

- Computer A: 2GHz clock, 10s CPU time to run a program.
- Build Computer B
 - Aim for 6s CPU time to run the same program
 - Can do a faster clock, but it requires 1.2 times as many clock cycles as computer A
($\text{Clock Cycles}_B = 1.2 \times \text{Clock Cycles}_A$)
- **Question: What clock rate should we tell the designer to target on Computer B?**



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- **Question: What clock rate should we tell the designer to target on Computer B?**

$$\begin{aligned}\text{Clock Rate}_B &= \frac{\text{Clock Cycles}_B}{\text{CPU Time}_B} \\ \text{Clock Cycles}_A &= \text{CPU Time}_A \times \text{Clock Rate}_A \\ &= 10\text{s} \times 2\text{GHz} = 20 \times 10^9 \\ \text{Clock Rate}_B &= \frac{1.2 \times 20 \times 10^9}{6\text{s}} = \frac{24 \times 10^9}{6\text{s}} = 4\text{GHz}\end{aligned}$$

- The computer needs to execute the instructions to run the program, and the execution time should depend on the number of instructions in a program
- **Clock cycles per instruction (CPI)**: the average number of clock cycles each instruction takes to execute
- The number of clock cycles required for a program is

$$\text{CPU Clock Cycles} = \text{Instruction Count} \times \text{CPI}$$

- The instruction count for a program
 - Determined by program, compiler, etc.
- CPI
 - Determined by how you design the CPU

```
hanoi:    addi $a0, $a0, -1
          bne  $a0, $zero, hanoi_1
          addi $v0, $zero, 1
          j    return
hanoi_1:  jal  hanoi
          sll  $v0, $v0, 1
          addi $v0, $v0, 1
return:   jr   $ra
```



The Classic CPU Performance Equation

$$\begin{aligned}\text{CPU Time} &= \text{CPU Clock Cycles} \times \text{Clock Period} \\ &= \frac{\text{CPU Clock Cycles}}{\text{Clock Rate}}\end{aligned}$$

$$\text{CPU Clock Cycles} = \text{Instruction Count} \times \text{CPI}$$

Use **Instruction Count**, **CPI**, **Clock Period** or **Clock Rate** to describe the **CPU Time**

$$\begin{aligned}\text{CPU Time} &= \text{Instruction Count} \times \text{CPI} \times \text{Clock Period} \\ &= \frac{\text{Instruction Count} \times \text{CPI}}{\text{Clock Rate}}\end{aligned}$$



CPI Example

- Two computers
 - Computer A: Clock Period = 250ps, CPI = 2.0
 - Computer B: Clock Period = 500ps, CPI = 1.2
 - Same program X
- **Which computer is faster to run X, and by how much?**

$$\begin{aligned}\text{CPU Time}_A &= \text{Instruct. Count} \times \text{CPI}_A \times \text{Clock Period}_A \\ &= C \times 2.0 \times 250\text{ps} = C \times 500\text{ps}\end{aligned}$$

$$\begin{aligned}\text{CPU Time}_B &= \text{Instruct. Count} \times \text{CPI}_B \times \text{Clock Period}_B \\ &= C \times 1.2 \times 500\text{ps} = C \times 600\text{ps}\end{aligned}$$

$$\frac{\text{CPU Time}_B}{\text{CPU Time}_A} = \frac{C \times 600\text{ps}}{C \times 500\text{ps}} = 1.2$$

A is faster

A is 1.2 times as fast as B



- Typically different instruction classes take different numbers of cycles

$$\text{Clock Cycles} = \sum_{i=1}^n (\text{CPI}_i \times \text{Instruction Count}_i)$$

- **Average CPI**

$$\text{CPI} = \frac{\text{Clock Cycles}}{\text{Instruction Count}} = \sum_{i=1}^n \left(\text{CPI}_i \times \frac{\text{Instruction Count}_i}{\text{Instruction Count}} \right)$$



Exercise

- A compiler designer is trying to decide between two instruction sequences for a particular computer. The CPI for each instruction class and the instruction counts (IC) for each instruction class are given as

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

- Question: What is the average CPI for each sequence?**

Exercise

- A compiler designer is trying to decide between two instruction sequences for a particular computer. The CPI for each instruction class and the instruction counts (IC) for each instruction class are given as

Class	A	B	C
CPI for class	1	2	3
IC in sequence 1	2	1	2
IC in sequence 2	4	1	1

- **Question: What is the average CPI for each sequence?**
 - Sequence 1: instruction count = 5
 - Clock Cycles
 $= 1 \times 2 + 2 \times 1 + 3 \times 2$
 $= 10$
 - Avg. CPI = $10/5 = 2.0$
 - Sequence 2: instruction count = 6
 - Clock Cycles
 $= 1 \times 4 + 2 \times 1 + 3 \times 1$
 $= 9$
 - Avg. CPI = $9/6 = 1.5$



Performance Summary

- The basic components of performance and how each is measured

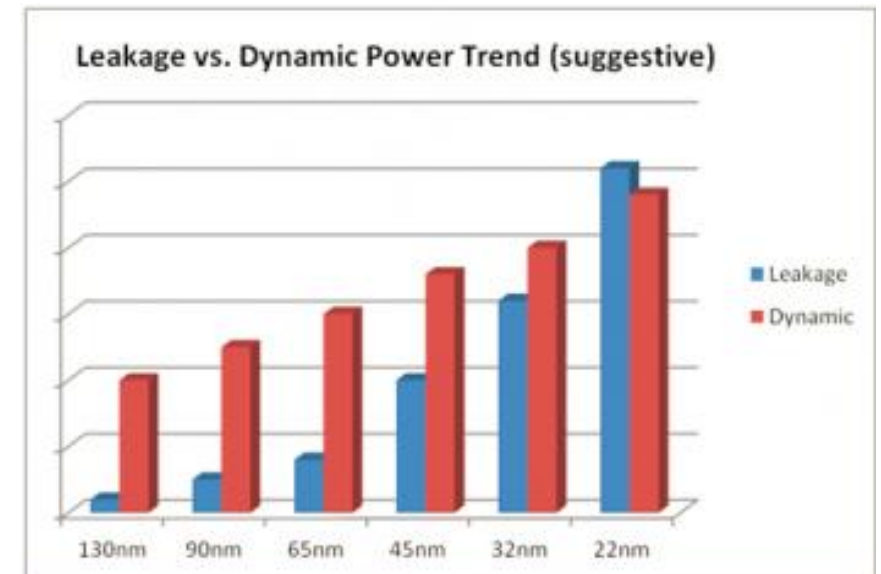
Components of performance	Units of measure
CPU execution time for a program	Seconds
Instruction count	Instructions executed for the program
Clock cycles per instruction (CPI)	Average number of clock cycles per instruction
Clock cycle time (period)	Seconds per clock cycle

- The big picture

$$\text{CPU Time} = \text{Instruction Count} \times \text{CPI} \times \text{Clock Period}$$

Power & Energy

- Dynamic power: $P = 0.5 * CV^2f$
 - C: effective capacitance
 - V: voltage
 - f: frequency, usually linear with V
- Doubling the clock frequency consumes more power than a quadcore processor!
- Static/Leakage power becomes the dominant factor in the most advanced process technologies.
- Power is the direct contributor of “heat”
 - Packaging of the chip
 - Heat dissipation cost
- Dynamic energy = $P * t$
 - Battery life is related to energy
 - Lower power does not necessarily mean better battery life





Bandwidth

- The amount of work (or data) during a period of time
 - Network or Disks: MB/sec, GB/sec, Gbps, Mbps
 - Game or Video: Frames per second
- Also called “throughput”, but with subtle differences
- “Work done” / “execution time”

Reliability

- Mean time to failure (MTTF)
 - Average time before a system stops working
 - Very complicated to calculate for complex systems
- Hardware can fail because of
 - Electromigration
 - Temperature
 - High-energy particle strikes



- Concepts of and basic factors that affect response time and bandwidth.
- Relative performance that is used to compare performance of different computers.
- Concepts and calculation of CPU time related factors.
- Concepts and calculation of CPI related factors.
- Knowledge of other performance metrics.



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Stay Tuned.